

Problem

A 110-V rms, 60-Hz source is applied to a load impedance $\mathbf{Z}$. The apparent power entering the load is 120 VA at a power factor of 0.707 lagging.

(a) Calculate the complex power.

(b) Find the rms current supplied to the load.

(c) Determine $\mathbf{Z}$.

(d) Assuming that $\mathbf{Z} = R + j\omega L$, find the values of R and L .

Step-by-step solution

Step 1 of 4

(a)

Calculate the power angle as follows:

Power factor is,

$$\cos \theta = 0.707$$

$$\begin{aligned}\theta &= \cos^{-1}(0.707) \\ &= 45^\circ\end{aligned}$$

Calculate the complex power.

$$S = S(\cos \theta) + jS(\sin \theta)$$

Substitute 45° for θ , 120 VA for S in equation.

$$\begin{aligned}S &= (120 \text{ VA})(\cos 45^\circ) + j(120 \text{ VA})(\sin 45^\circ) \\ &= 84.84 + j84.84 \text{ VA}\end{aligned}$$

Therefore, the value of complex power is $\boxed{84.84 + j84.84 \text{ VA}}$.

Step 2 of 4

(b)

Calculate the rms current supplied to the load.

$$I_{\text{rms}} = \frac{S}{V_{\text{rms}}}$$

Substitute 120 VA for S , 110 V for V_{rms} in equation.

$$\begin{aligned}I_{\text{rms}} &= \frac{120 \text{ VA}}{110 \text{ V}} \\ &= 1.09 \text{ A rms}\end{aligned}$$

Therefore, the value of rms current supplied to the load is $\boxed{1.09 \text{ A rms}}$.

[Comments \(1\)](#)



Anonymous

My initial thought was to plug in $S = (84.84 + j84.84) \text{ VA}$ into the rms current equation...why don't we do that?

(c)

Calculate the value of impedance.

$$\mathbf{Z} = \frac{\mathbf{S}}{I_{rms}^2}$$

Substitute the values $84.84 + j84.84 \text{ VA}$ for $\mathbf{S}$ and 1.09 A rms for I_{rms} equation.

$$\begin{aligned}\mathbf{Z} &= \frac{84.84 + j84.84 \text{ VA}}{(1.09 \text{ A rms})^2} \\ &= \frac{84.84 + j84.84 \text{ VA}}{1.1881} \\ &= 71.4 + j71.4 \Omega\end{aligned}$$

Therefore, the value of impedance is $\boxed{71.4 + j71.4 \Omega}$.

(d)

Consider the impedance, $\mathbf{Z} = R + j\omega L$ Compare impedance expression with $\mathbf{Z} = 71.4 + j71.4 \Omega$.

$$R = 71.4 \Omega$$

$$\omega L = 71.4 \Omega$$

$$2\pi fL = 71.4 \Omega$$

$$L = \frac{71.4 \Omega}{2\pi f}$$

Substitute 60 Hz for f in equation.

$$\begin{aligned}L &= \frac{71.4 \Omega}{2\pi(60 \text{ Hz})} \\ &= 189.49 \text{ mH}\end{aligned}$$

Therefore, the value of resistance is $\boxed{71.4 \Omega}$ and the value of inductance is $\boxed{189.49 \text{ mH}}$.